

Python if-else Statement

Python if else Statement

The if-else statement in **Python** is used to execute a block of code when the condition in the **if statement** is true, and another block of code when the condition is false.

Syntax of if-else Statement

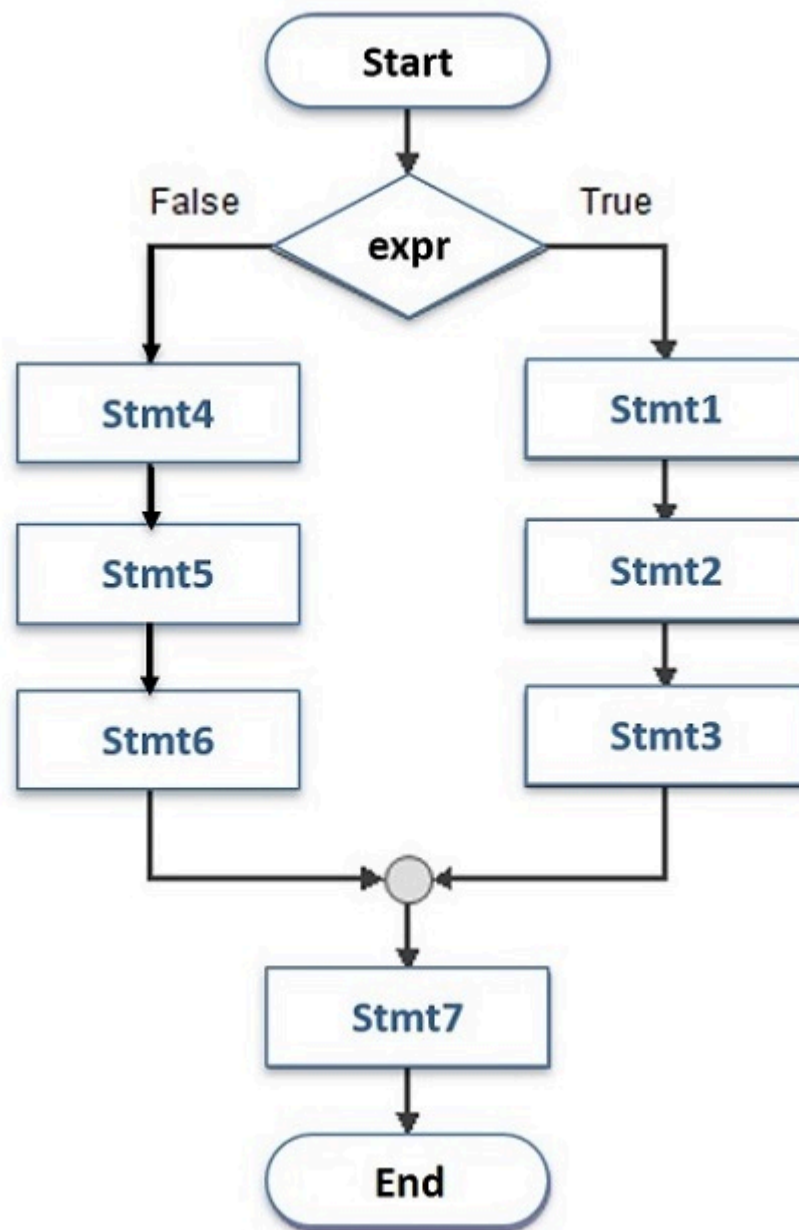
The syntax of an if-else statement in Python is as follows –

```
if boolean_expression:
    # code block to be executed
    # when boolean_expression is true
else:
    # code block to be executed
    # when boolean_expression is false
```

If the boolean expression evaluates to TRUE, then the statement(s) inside the **if block** will be executed otherwise statements of the **else block** will be executed.

Flowchart of if-else Statement

This flowchart shows how if-else statement is used –



If the **expr** is True, block of stmt1, 2, 3 is executed then the default flow continues with stmt7. However, if the **expr** is False, block stmt4, 5, 6 runs then the default flow continues.

Python implementation of the above flowchart is as follows –

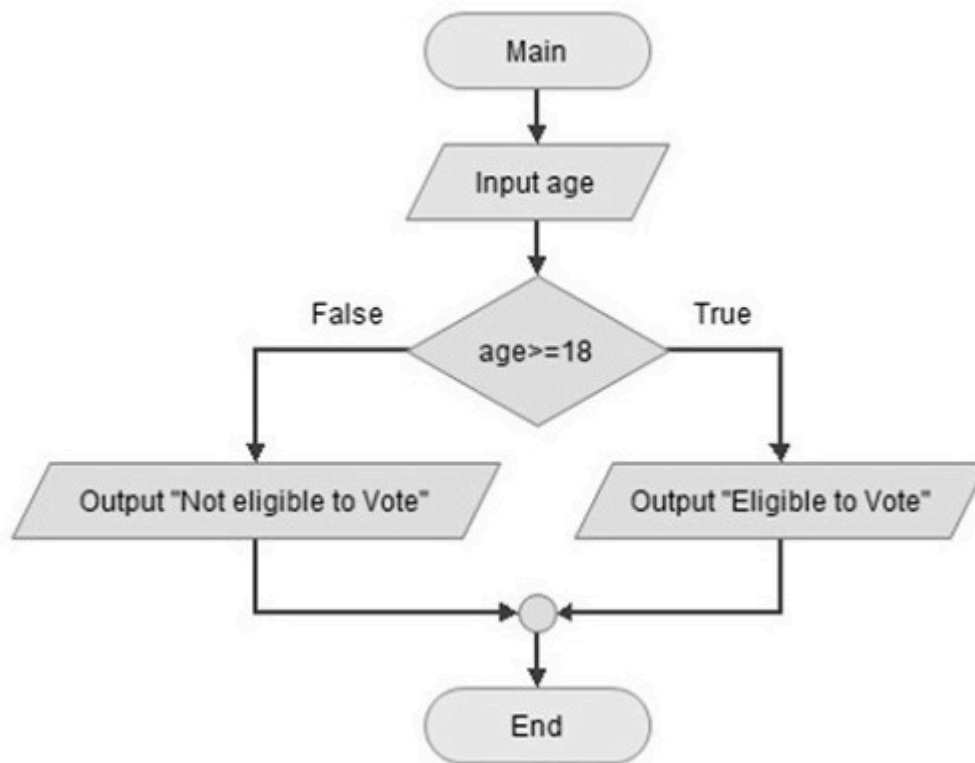
```

if expr==True:
    stmt1
    stmt2
    stmt3
else:
    stmt4
    stmt5
    stmt6
stmt7

```

Python if-else Statement Example

Let us understand the use of **if-else** statements with the following example. Here, variable **age** can take different values. If the expression **age > 18** is true, then **eligible to vote** message will be displayed otherwise **not eligible to vote** message will be displayed. Following flowchart illustrates this logic –



Now, let's see the Python implementation the above flowchart.

```
age=25
print ("age: ", age)
if age >=18:
    print ("eligible to vote")
else:
    print ("not eligible to vote")
```

On executing this code, you will get the following **output** –

```
age: 25
eligible to vote
```

To test the **else block**, change the **age** to 12, and run the code again.

```
age: 12
not eligible to vote
```

Python if elif else Statement

The **if elif else** statement allows you to check multiple expressions for TRUE and execute a block of code as soon as one of the conditions evaluates to TRUE.

Similar to the **else block**, the **elif block** is also optional. However, a program can contain only one **else block** whereas there can be an arbitrary number of **elif blocks** following an **if block**.

Syntax of Python if elif else Statement

```
if expression1:
    statement(s)
elif expression2:
    statement(s)
elif expression3:
    statement(s)
else:
    statement(s)
```

How if elif else Works?

The keyword **elif** is a short form of **else if**. It allows the logic to be arranged in a cascade of **elif** statements after the first **if** statement. If the first **if** statement evaluates to false, subsequent **elif** statements are evaluated one by one and comes out of the cascade if any one is satisfied.

Last in the cascade is the **else** block which will come in picture when all preceding if/elif conditions fails.

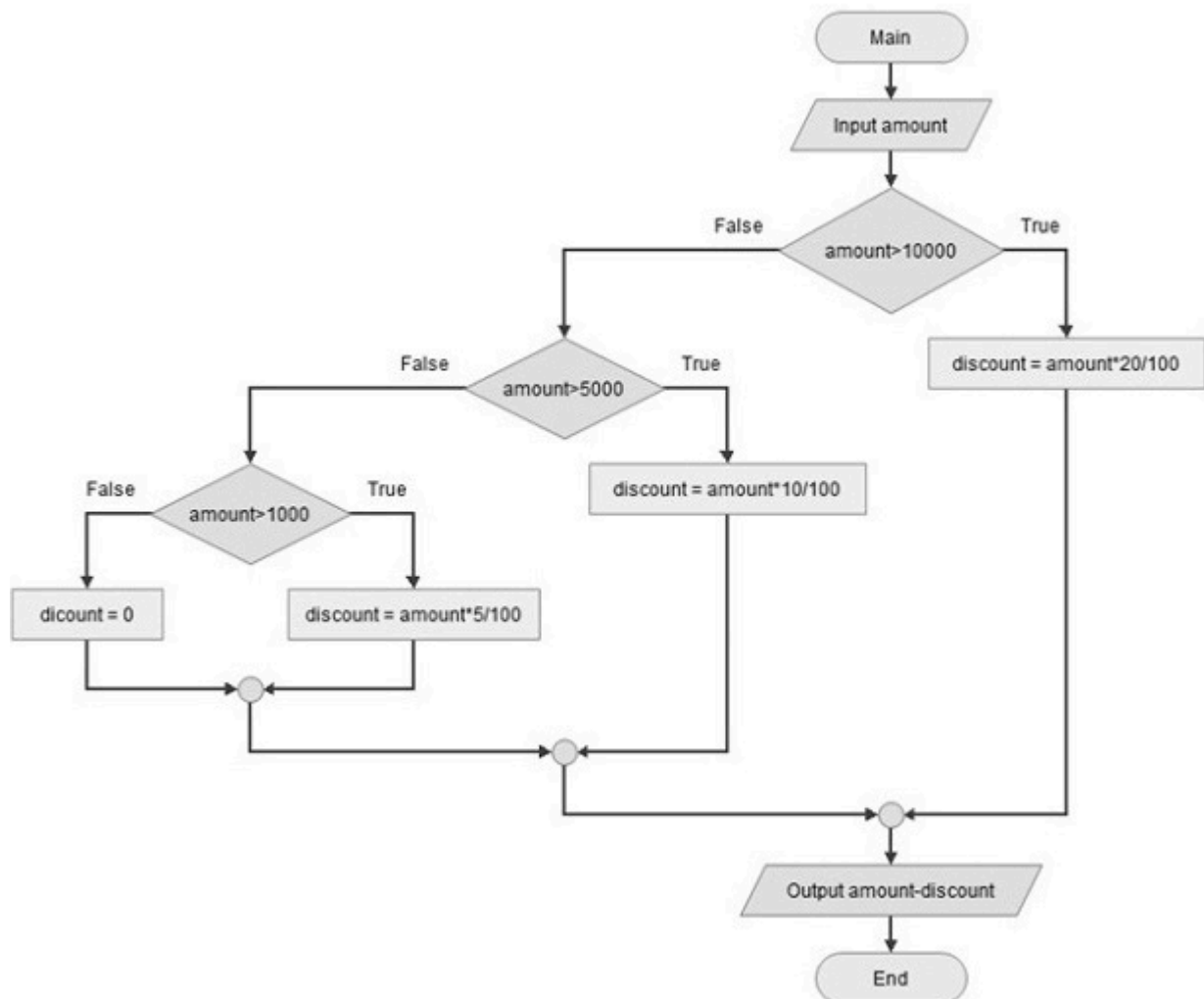
Example

Suppose there are different slabs of discount on a purchase –

- 20% on amount exceeding 10000,
- 10% for amount between 5-10000,

- 5% if it is between 1 to 5000.
- no discount if amount < 1000

The following flowchart illustrates these conditions –



We can write a Python code for the above logic with **if-else** statements –

```

amount = 2500
print('Amount = ', amount)
if amount > 10000:
    discount = amount * 20 / 100
else:
    if amount > 5000:
        discount = amount * 10 / 100
    else:
        if amount > 1000:
            discount = amount * 5 / 100

```

```

        else:
            discount = 0

print('Payable amount = ',amount - discount)

```

Set **amount** to test all possible conditions: 800, 2500, 7500 and 15000. The outputs will vary accordingly –

```

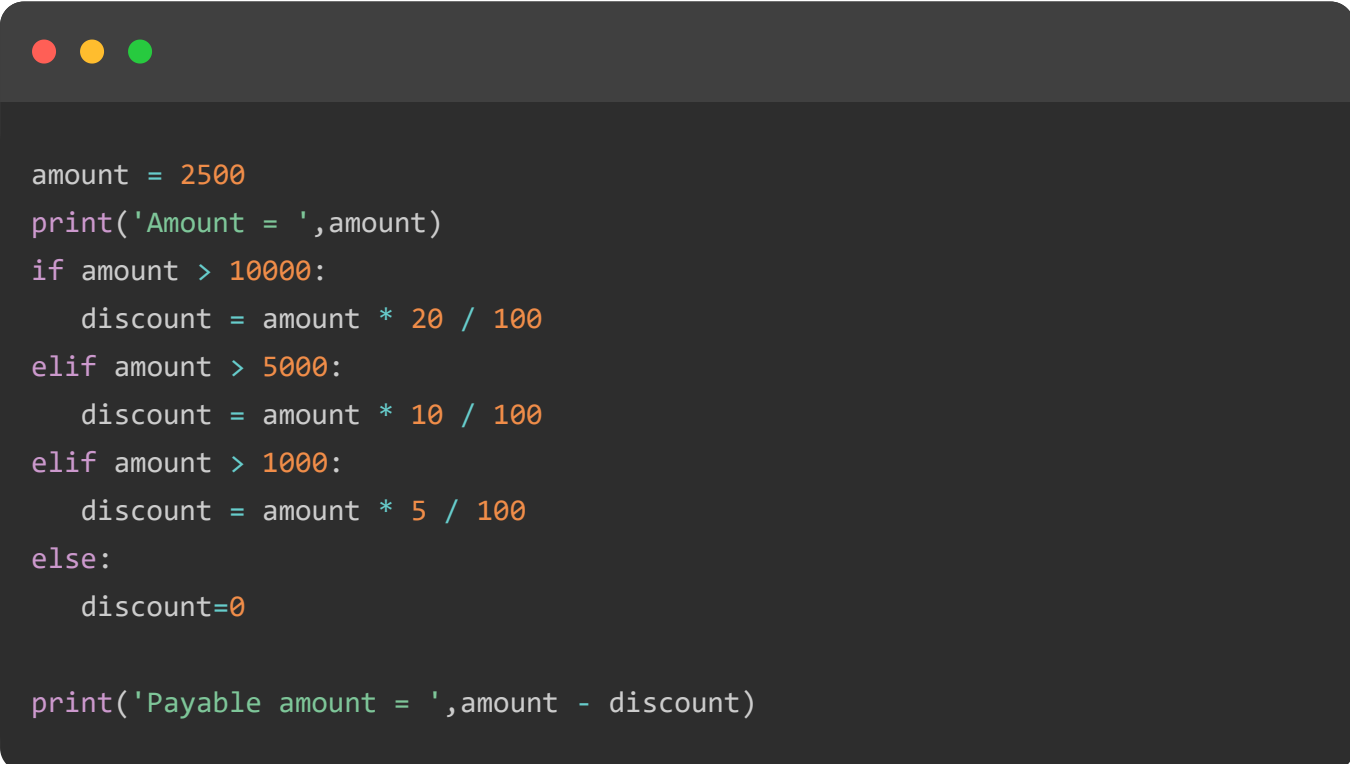
Amount: 800
Payable amount = 800
Amount: 2500
Payable amount = 2375.0
Amount: 7500
Payable amount = 6750.0
Amount: 15000
Payable amount = 12000.0

```

While the code will work perfectly fine, if you look at the increasing level of indentation at each if and else statement, it will become difficult to manage if there are still more conditions.

Python if elif else Statement Example

The **elif statement** makes the code easy to read and comprehend. Following is the Python code for the same logic with if elif else statements –



```

amount = 2500
print('Amount = ',amount)
if amount > 10000:
    discount = amount * 20 / 100
elif amount > 5000:
    discount = amount * 10 / 100
elif amount > 1000:
    discount = amount * 5 / 100
else:
    discount=0

print('Payable amount = ',amount - discount)

```

The output of the above code is as follows –

```
Amount: 2500
Payable amount = 2375.0
```

TOP TUTORIALS

Python Tutorial
Java Tutorial
C++ Tutorial
C Programming Tutorial
C# Tutorial
PHP Tutorial
R Tutorial
HTML Tutorial
CSS Tutorial
JavaScript Tutorial
SQL Tutorial

TRENDING TECHNOLOGIES

Cloud Computing Tutorial
Amazon Web Services Tutorial
Microsoft Azure Tutorial
Git Tutorial
Ethical Hacking Tutorial
Docker Tutorial
Kubernetes Tutorial
DSA Tutorial
Spring Boot Tutorial
SDLC Tutorial
Unix Tutorial

CERTIFICATIONS

Business Analytics Certification

Java & Spring Boot Advanced Certification
Data Science Advanced Certification
Cloud Computing And DevOps
Advanced Certification In Business Analytics
Artificial Intelligence And Machine Learning
DevOps Certification
Game Development Certification
Front-End Developer Certification
AWS Certification Training
Python Programming Certification

COMPILERS & EDITORS

Online Java Compiler
Online Python Compiler
Online Go Compiler
Online C Compiler
Online C++ Compiler
Online C# Compiler



Chapters ▾

Categories ≡

Online Bash Terminal
Online SQL Compiler
Online Html Editor

[ABOUT US](#) | [OUR TEAM](#) | [CAREERS](#) | [JOBS](#) | [CONTACT US](#) | [TERMS OF USE](#) |
[PRIVACY POLICY](#) | [REFUND POLICY](#) | [COOKIES POLICY](#) | [FAQ'S](#)



Tutorials Point is a leading Ed Tech company striving to provide the best learning material on technical and non-technical subjects.

© Copyright 2026. All Rights Reserved.